

Abhay Goudannavar

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PROFESSIONAL SUMMARY

AI/ML undergraduate and active open-source contributor with a track record of building features and optimizing code for major AI research frameworks (including NVIDIA and Google DeepMind). Proficient in Python and full-stack development, with strong experience navigating complex codebases, executing performance optimizations, and collaborating through rigorous PR reviews.

EDUCATION

Medhavi Skills University

B.Tech in Artificial Intelligence and Machine Learning

Mahesh PU College, Bengaluru

Pre-University

TECHNICAL SKILLS

Programming Languages:	Python, JavaScript, C++, Java, Rust (Basics), Dart
Machine Learning & Data Science:	PyTorch, NumPy, Pandas, Jupyter Notebook
Web & App Development:	Vue.js, React, HTML/CSS, TypeScript (Basics), Flutter
Backend & Cloud Services:	FastAPI, Firebase
DevOps & Tools:	Git, GitHub, Docker, Linux/WSL, Key Hole Mapping

OPEN SOURCE EXPERIENCE

Open Source Contributor — NVIDIA (earth2studio)

- Implemented `torch.compile` support for autoregressive inference rollouts, achieving 1.5x to 3x speedups via CUDA graph optimization with safe eager-mode fallbacks.
- Developed a pure PyTorch multivariate Energy Score metric for ensemble forecast verification to evaluate spatial correlations accurately across grid points.

Open Source Contributor — mllam (neural-lam)

- Stabilized multi-step autoregressive model training by integrating configurable gradient clipping into the PyTorch Lightning pipeline to prevent exploding gradients.
- Optimized GPU data transfer throughput by enabling pinned memory in DataLoaders for asynchronous CPU-to-GPU Direct Memory Access (DMA).

Open Source Contributor — Google DeepMind (OpenSpiel)

- Configured and optimized the Python API for OpenSpiel to support complex game-theoretic and reinforcement learning environments.
- Streamlined multi-agent game simulation workflows, improving the foundation for advanced AI research tooling.

Open Source Contributor — RUXAILAB

- Contributed features and UI improvements to a production Vue.js and Firebase codebase.
- Implemented camera-based profile photo capture using browser MediaDevices APIs.
- Fixed runtime issues and improved system stability by resolving complex logout cleanup errors.
- Navigated rigorous PR reviews, resolved merge conflicts, and ensured CI quality checks (SonarCloud) passed successfully.

PROJECTS

Sustainable Food Tracker

- Developed a health and sustainability application that tracks carbohydrates, protein, and other essential nutrients instantly by scanning packaged food barcodes.
- Engineered a gamified "sustainability score" system, allowing users to log healthy eating habits and compete on a leaderboard with friends and family.

Nexora - Educational Platform

- Built a comprehensive educational platform designed to organize study materials, textbooks, and syllabus details for upcoming examinations.
- Integrated a dynamic recommendation feature to suggest relevant educational YouTube tutorials based on specific exam syllabi and application dates.